

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-8263
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Architecture - Import Data Using Transform Maps

Layer / Component	Flow
Data Source	Spreadsheet containing employee information
Import Layer	ServiceNow Import Set loads records into the Employee Import staging table
Transformation Layer	Sample Spreadsheet Import Transform Map maps source fields to target fields
Data Layer	Employee Test table stores the transformed employee records
Reporting Layer	Employee Analytics Dashboard presents employee information by department and location

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	ServiceNow Web UI
2	Application Logic-1	Logic for a process in the application	ServiceNow Import Sets
3	Application Logic-2	Logic for a process in the application	ServiceNow Transform Maps
4	Application Logic-3	Logic for a process in the application	Transform Map Coalesce using

			Employee ID
5	Database	Data Type, Configurations etc.	ServiceNow platform database
6	Cloud Database	Database Service on Cloud	ServiceNow-managed cloud database
7	File Storage	File storage requirements	Spreadsheet / Local Filesystem
8	External API-1	Purpose of External API used in the application	Not used
9	External API-2	Purpose of External API used in the application	Not used
10	Machine Learning Model	Purpose of Machine Learning Model	Not used
11	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration:	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used	No separate open-source framework; ServiceNow platform
2	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	ServiceNow Roles and ACLs
3	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Import Sets and Transform Maps support batch employee-data processing
4	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	ServiceNow Cloud Platform
5	Performance	Design consideration for the performance of the application (number of requests per sec,	Bulk import and transformation reduce manual record-by-record data entry

		use of Cache, use of CDN's) etc.	
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References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>